



ChildSafe

by InnoSafe™

“Keeping a child in safe zone”

Every second counts. For parents of active children, a single moment of distraction in a public space can lead to accidents, injury, or abduction. The consequences are devastating and often irreversible. **ChildSafe** is built to change that, giving parents real-time peace of mind and keeping every child within reach, no matter where life takes them.

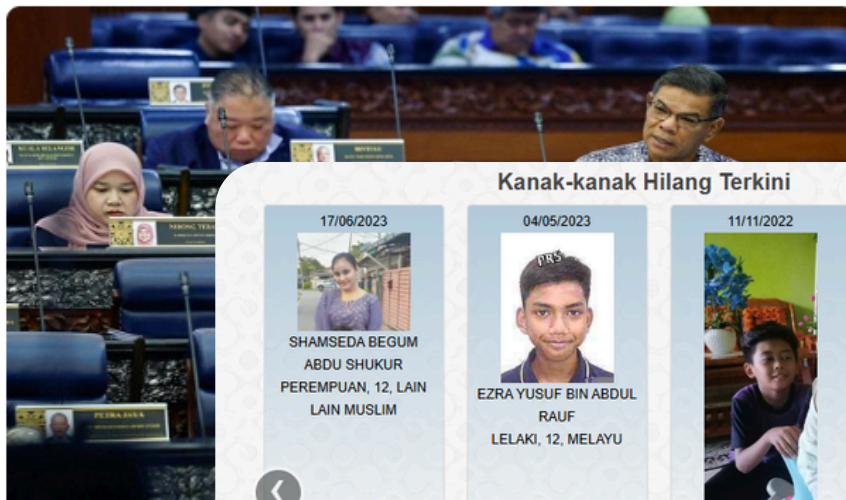


THE MISSING CHILDREN CRISIS



96 peratus daripada 3,847 kanak-kanak hilang berjaya ditemui

17/10/2024 | 11:59 AM - NASIONAL



Kanak-kanak Hilang Terkini



3,847

Children lost in Malaysia between 2020 and 2024

770

Average child lost in a year.

Source:

<https://berita.rtm.gov.my/nasional/senarai-berita-nasional/senarai-artikel/96-peratus-daripada-3-847-kanak-kanak-hilang-berjaya-ditemui/>

Private & Confidential

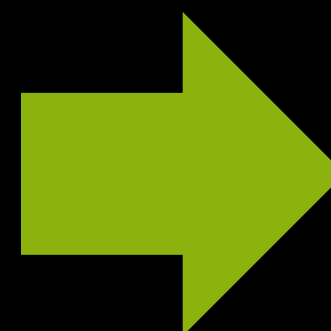


What if your phone **screamed at you** before your child was lost — not after?

ChildSafe prevents. It defines a boundary — and the moment that boundary is crossed, the parent knows. Instantly.

Current State

React. Recover. Regret.



ChildSafe

Detect. Alert. Prevent.



It Keeps Your Child Inside the Fence.



Proactive, Real-Time & Scalable Protection

ChildSafe is a wearable BLE beacon worn by the child, paired with a parent smartphone app. The parent draws a safe zone radius. The moment the child exits — the phone alerts immediately. No GPS. No monthly fees. No signal dependency.



Micro-Location Tracking Engine (BLE & RSSI)



Alert Notification Module



Affordable Child Safety System

**WORKS
OUTDOORS**

INSTANT ALERT

LOW POWER

NO GPS NEEDED



HOW CHILDSAFE WORKS



“Keeping a child in safe zone”

HOW BIG IS THE OPPORTUNITY?



Target Audience User Profile

DEMOGRAPHICS

- **Age:** 25 - 45 years old.
- **Urban:** M40 & T20 corporate parents.
- **Rural:** B40 & M40 local working class.

GEOGRAPHICS

- **Urban:** All cities in Malaysia
- **Rural:** Sub-urban towns & active rural villages.

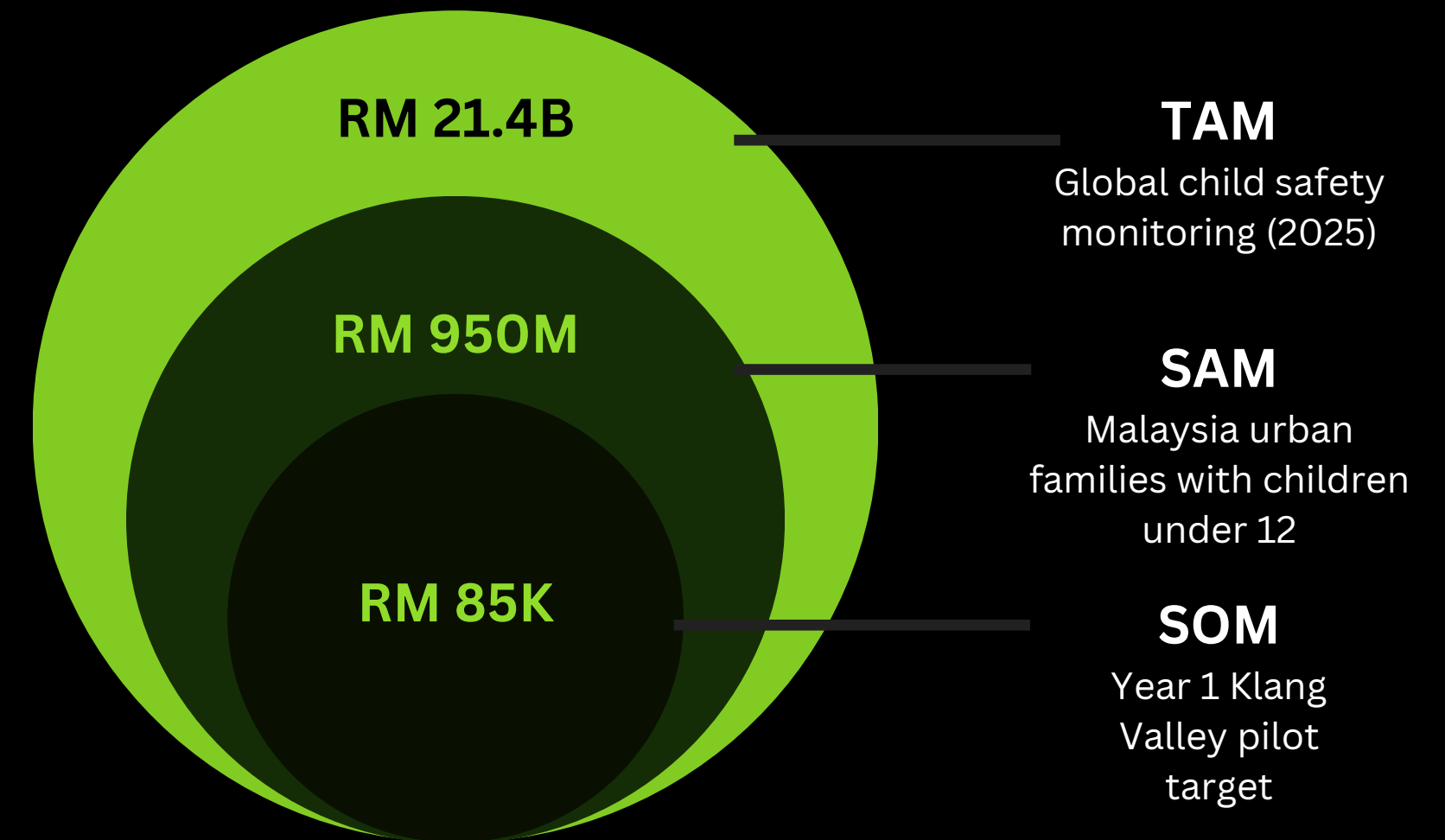
BEHAVIOR

- **Urban:** Visits crowded malls & urban parks.
- **Rural:** Visits night markets & open-perimeter spaces.

PSYCHOGRAPHICS

- **Urban:** Anxiety over crowded space separation.
- **Rural:** Anxiety over wandering into nearby hazards.

Market Size Projection





“OUR ASK: **RM500,000**”

Marketing & Customer Acquisition

- RM 100, 000
- B2B outreach (malls, transits), branding, digital ads.

Product Development

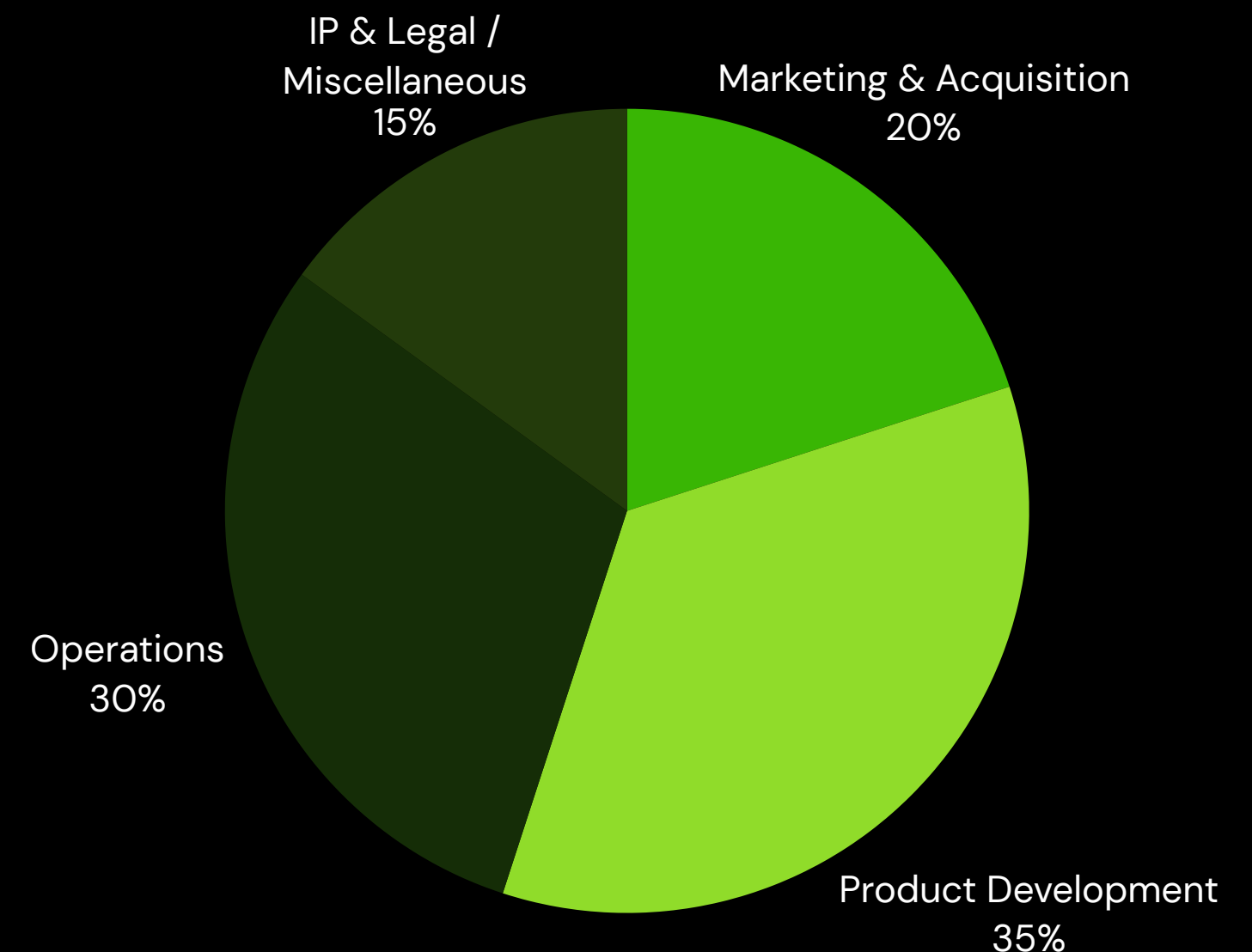
- RM 175, 000
- RSSI algorithm optimization & mobile app deployment
- Moving from Arduino prototype to custom ESP32 PCB.

Operations

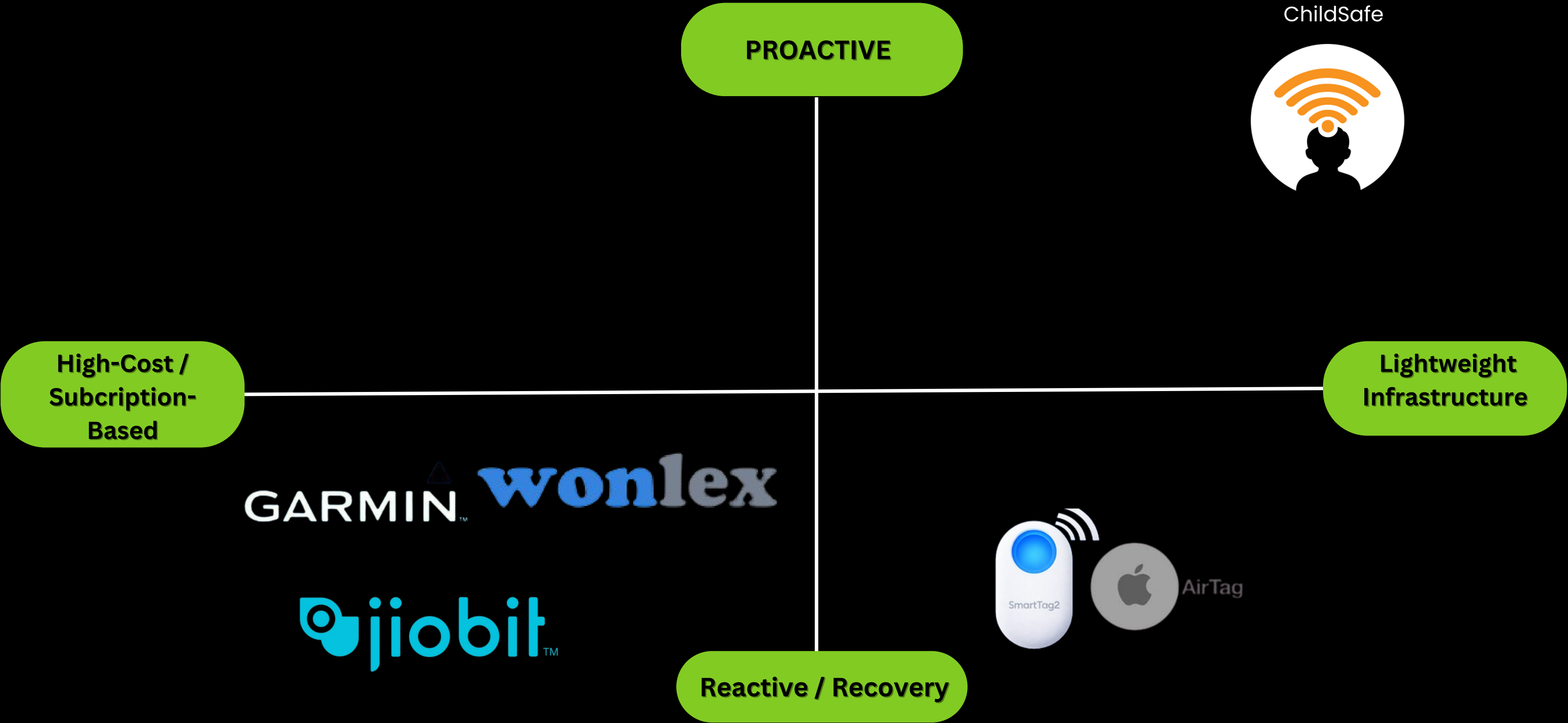
- RM 150,000
- Core engineering team, developer hires, supply chain setup.

IP & Legal /Miscellaneous

- RM 75,000
- RSSI algorithm patent filing, MCMC certification, buffer.



NOVELTY / UNIQUE SELLING POINT





CHILDSAFE IN 5 YEARS

YEAR 1

YEAR 2

YEAR 3

YEAR 4 - 5

LAUNCH

- Core BLE + RSSI hardware system finalized.
- Launch official tracking app for iOS & Android.
- Secure first 342 units adoptions through local mall pilots.

KNOWLEDGE HUB

- Introduce in-app “Parental Guidance & Knowledge” database.
- Daily child-care safety tips and emergency guidance alerts.
- Refine hardware battery life to 2x extended duration.

SMART SAFETY

- Integrate AI Smart Alert layers for multi-child monitoring.
- Launch B2B safety management packages for school transits.
- Onboard 100 childcare institutions to the safety fleet.

COMMERCIALIZE

- Realize market-ready product V3 for retail launch.
- Expand ASEAN regional market outreach for smart parenting.
- Partner with the public safety sector for B2G initiatives.

OUR TEAM



Founder & CEO

Harleiff bin Haniff

2 yrs of product design
& development
experience



Co-Founder

Ahmad Zikry bin Aini

2 years of embedded
systems experience



Chief Technology Officer

Wan Shafiq bin
Wan Ahmad Sufian

2 years of software
development
experience



Marketing Strategy

Muhammad Ezam
bin Fauzi

2 years marketing and
graphic design
experience



TECHNICAL PART





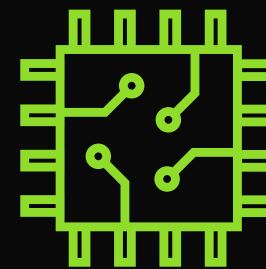
TECHNICAL OVERVIEW

Deep-dive into the proprietary stack enabling meter-perfect **indoor tracking**.



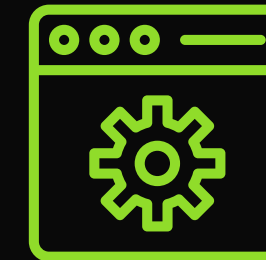
ARCHITECTURE

Low-latency BLE communication between the dynamic device and point reference.



ALGORITHM

Dynamic data filtering to prevent false threshold trigger.

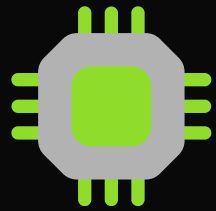


MINIATURIZATION

Roadmap minituarization of the prototype.



HARDWARE & SOFTWARE COMPONENTS



Hardware

Device A

- Arduino Uno
- Bluetooth Module HC-05
- Powerbank 20000mAh

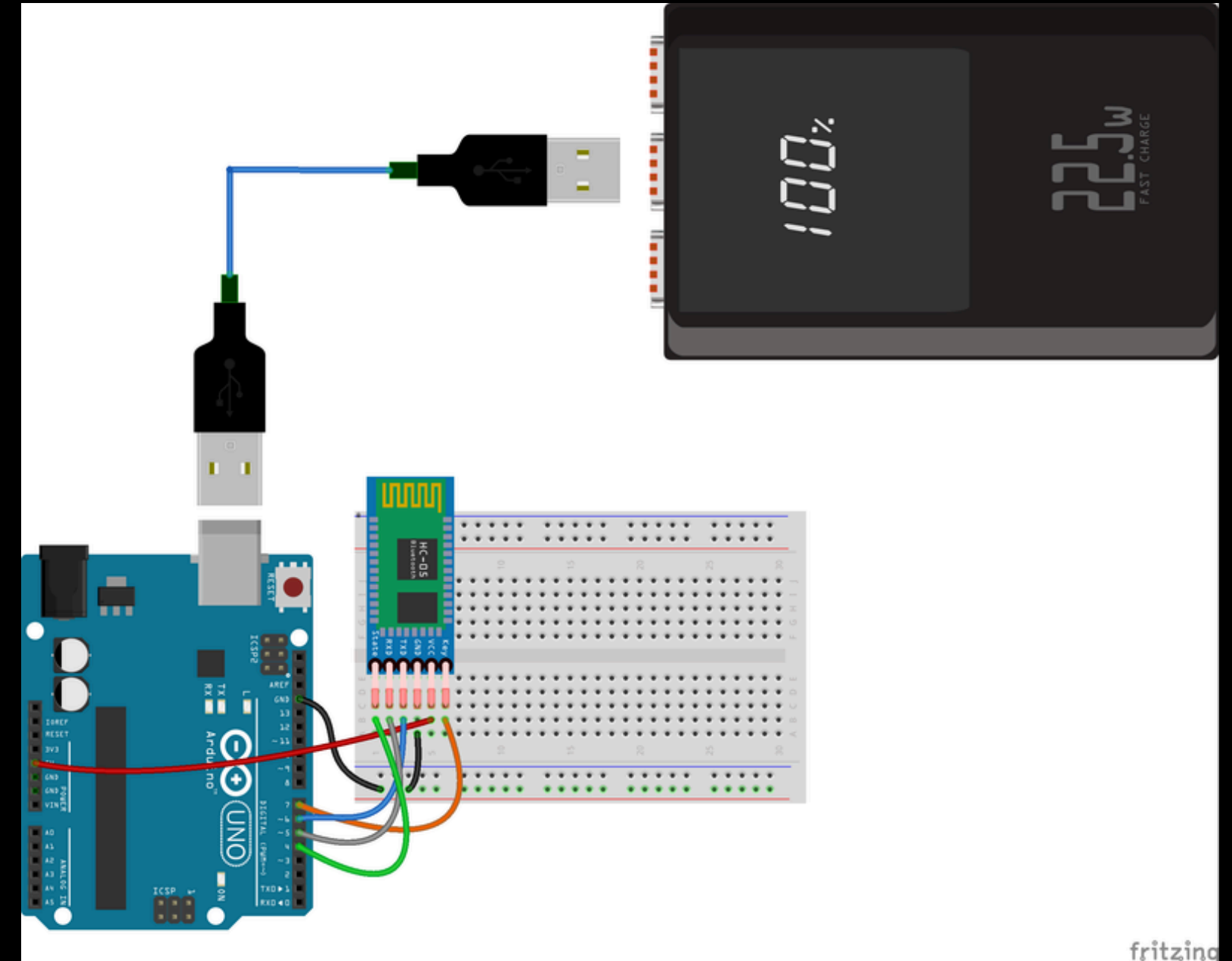
Device B

- Computer
- Bluetooth Dongle



Software

- Embedded C/C++
 - (Arduino IDE)
- HTML , CSS, JavaScript
 - (Web server)
- MySQL & SQL Language
 - (phpMyAdmin)





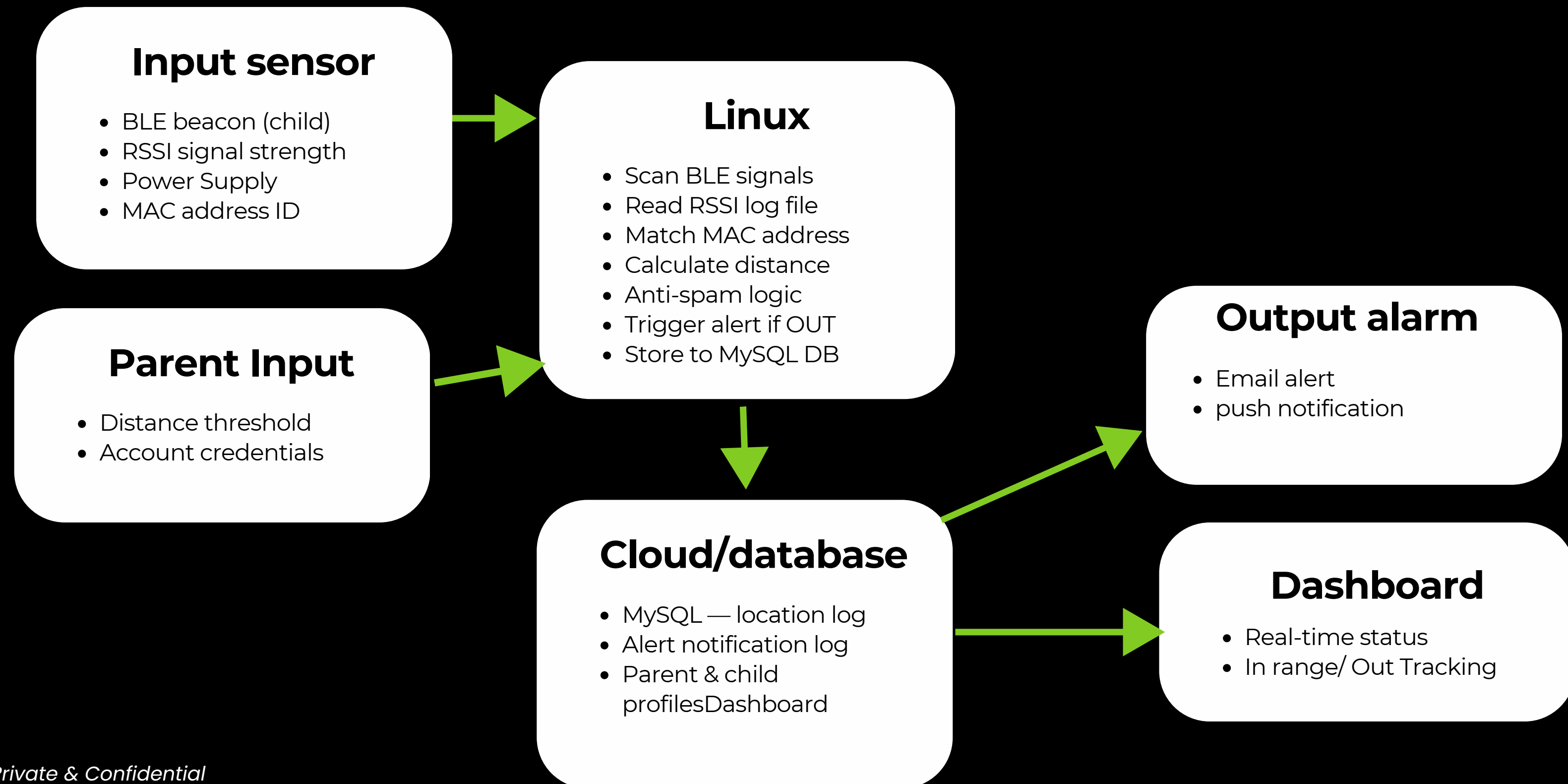
SYSTEM ARCHITECTURE

Signal Strength (RSSI) Zone Classification

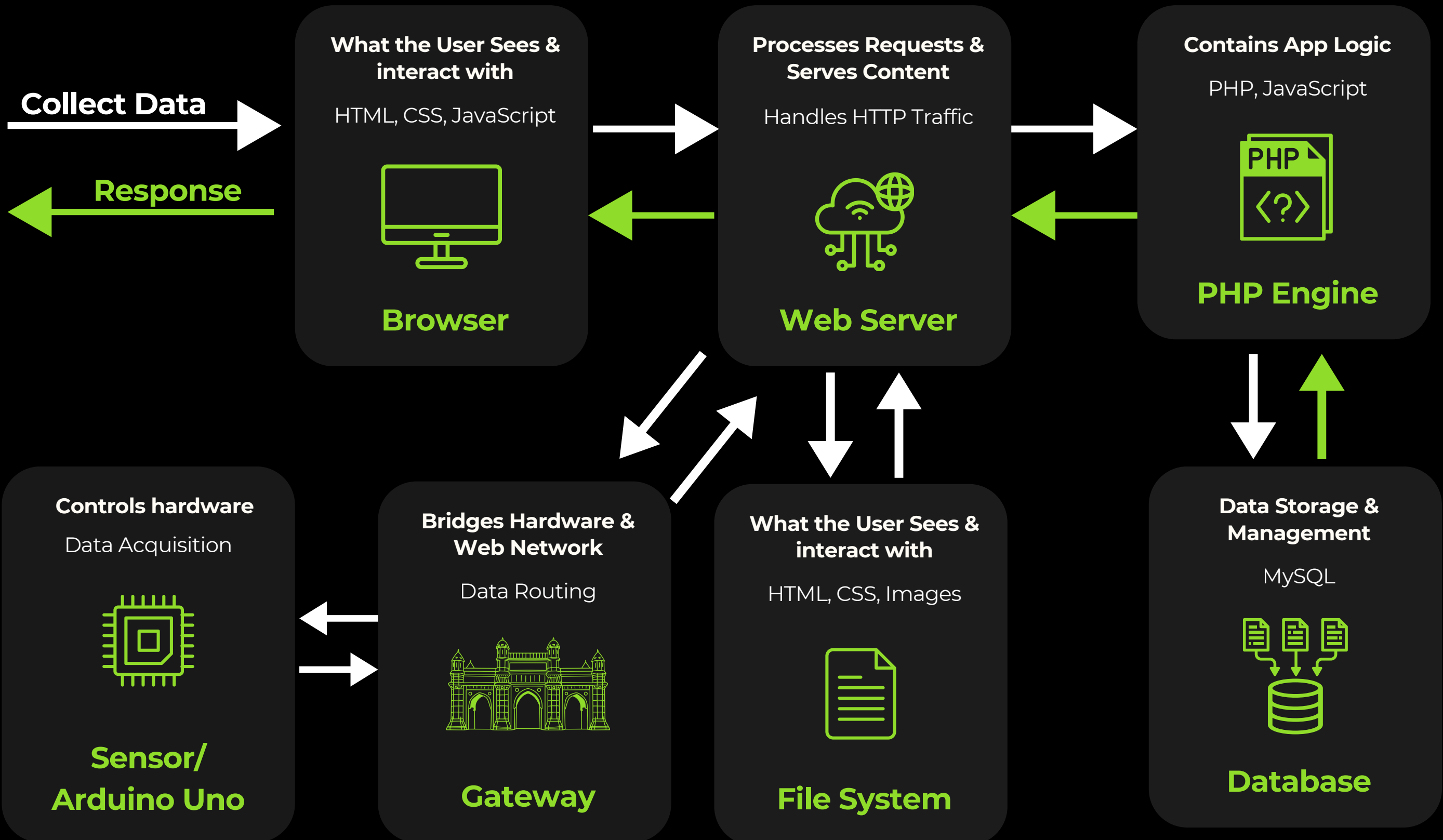
Zone Status	RSSI Range (Signal Strength)	System Interpretation & Actions
IN RANGE	0.0m until 2.5m	Target detected within safe perimeter.
WARNING	2.5m until 3.0m	Target approaching perimeter threshold
OUT OF RANGE	3m and above	Target lost or cleared perimeter.



HIGH LEVEL BLOCK DIAGRAM

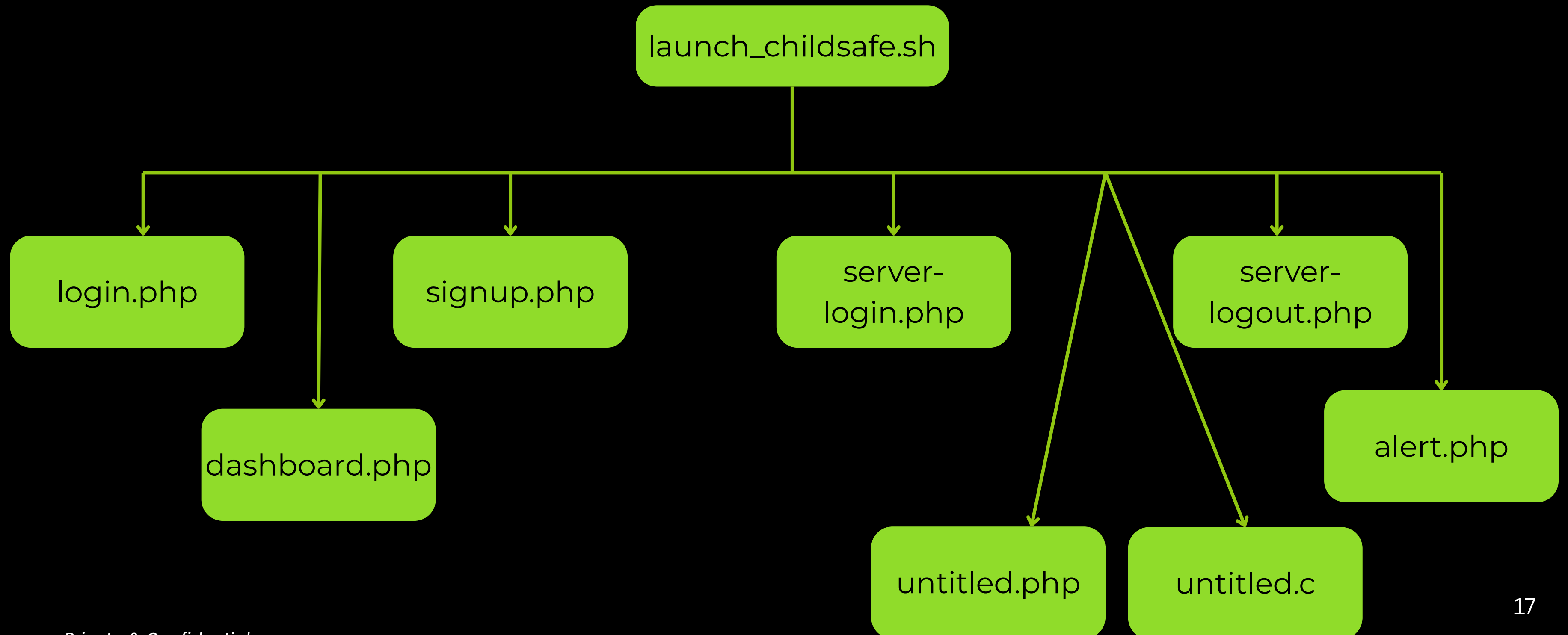


System Design and Architecture





FIRMWARE STRUCTURE





DATA FLOW & COMMUNICATION PROCESS

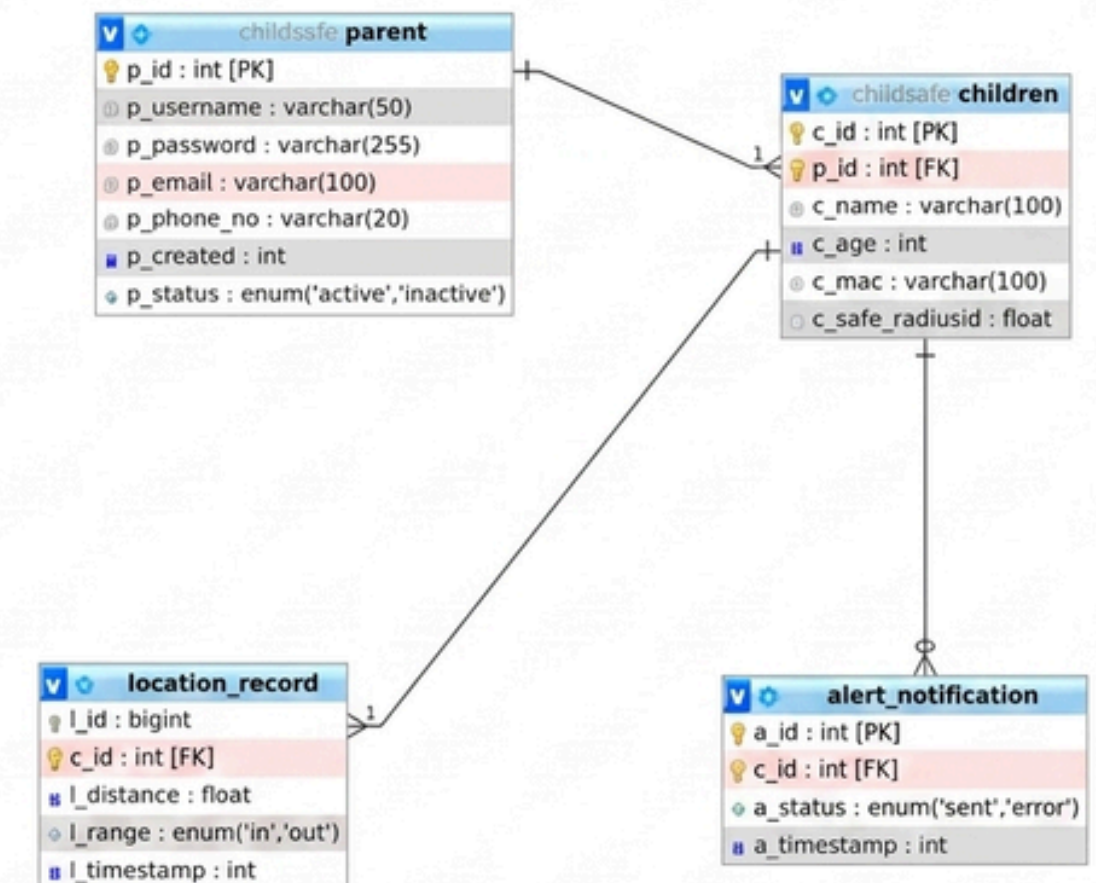
Parent Table

#	Name	Type	Collation	Attributes	Null	Default
1	p_id	int			No	None
2	p_username	varchar(50)	utf8mb4_0900_ai_ci		No	None
3	p_password	varchar(255)	utf8mb4_0900_ai_ci		No	None
4	p_email	varchar(50)	utf8mb4_0900_ai_ci		No	None
5	p_phone_no	varchar(14)	utf8mb4_0900_ai_ci		No	None
6	p_created	int			No	None
7	p_status	enum('active', 'inactive')	utf8mb4_0900_ai_ci		No	None

Location Table

#	Name	Type	Collation	Attributes	Null	Default
1	l_id	bigint			No	None
2	c_id	int			No	None
3	l_distance	float			No	None
4	l_range	enum('in', 'out')	utf8mb4_0900_ai_ci		No	None
5	l_timestamp	int			No	None

Data Scheme



Child Table

#	Name	Type	Collation	Attributes	Null	Default	Comm
1	c_id	int			No	None	
2	p_id	int			No	None	
3	c_name	varchar(100)	utf8mb4_0900_ai_ci		No	None	
4	c_age	int			No	None	
5	c_mac	varchar(50)	utf8mb4_0900_ai_ci		No	None	
6	c_safe_radius	float			No	None	

Alert Table

#	Name	Type	Collation	Attributes	Null	Default
1	a_id	int			No	None
2	c_id	int			No	None
3	a_status	enum('sent', 'error')	utf8mb4_0900_ai_ci		No	None
4	a_timestamp	int			No	None

DATABASE SYSTEM



Login interface



Welcome to **ChildSafe**

Sign in to your account below

Registration successful!

Username

Zikry

Password

••••

Login

Don't have an account yet? [Sign Up](#)

Home interface

ChildSafe

User Active

Logout

Hi EzamOfficial 🖐️

Add Child



Child Name

Adam

Age

15

Safe Radius

3 m

View History

1 m

In Range

Last update: 2026-05-21 18:46:26



Child Name

Hal

Age

9

Safe Radius

1 m

View History

No reading

No reading

Last update: No reading



Child Name

Age

Safe Radius

View History

No reading

No reading

Last update: No reading

Signup interface



Create your **ChildSafe** account

Fill in your details below

Username

Zikry

Password

••••

Email

zikry@gmail.com

Phone Number

0123776635

Sign Up

Already a member? [Sign In](#)

Database interface

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

SELECT * FROM `parent`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

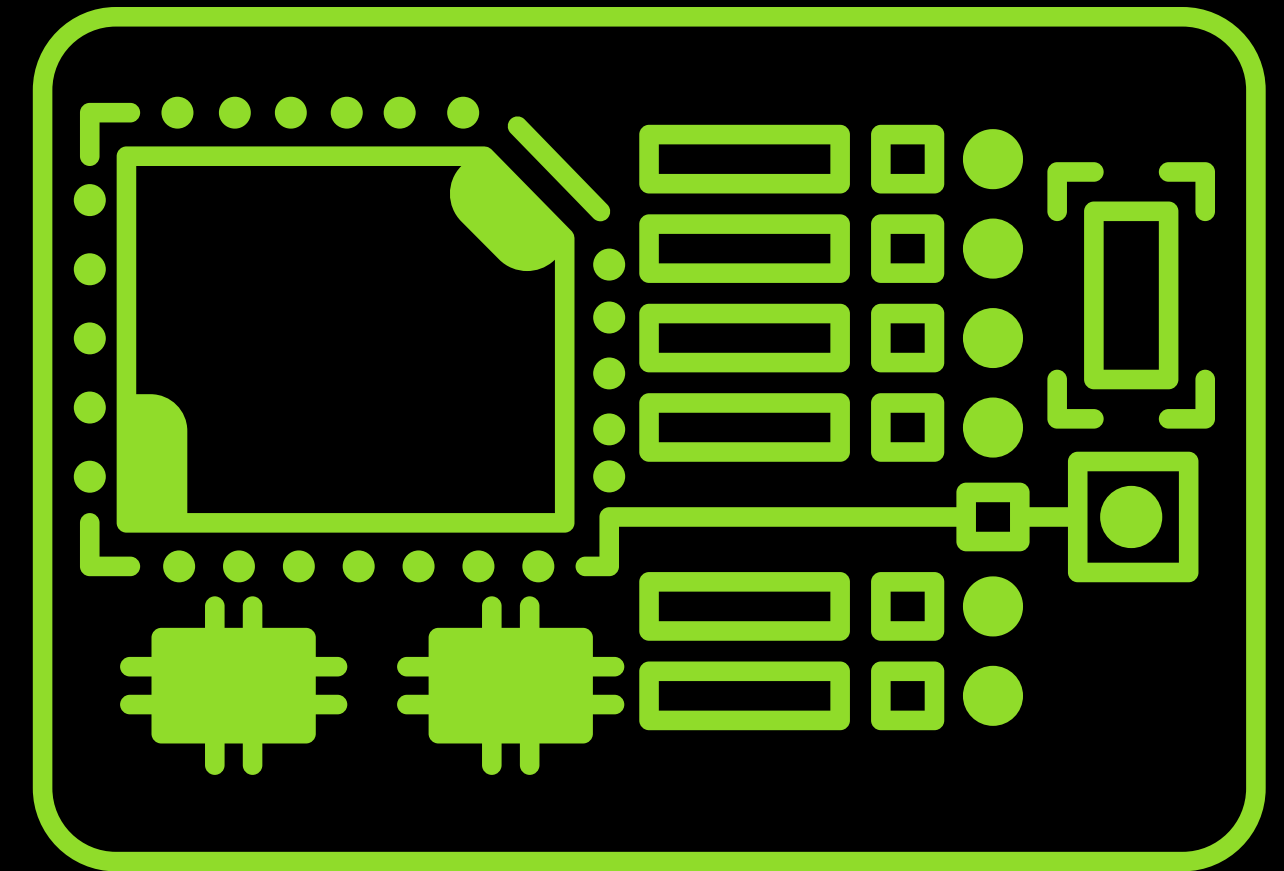
☐ Show all | Number of rows: 25 | Filter rows: Sort by key: None

+ Options

				p_id	p_username	p_password	p_email	p_phone_no	p_created	p_status
☐				1	EzamOfficial	\$2y\$10\$7XsESgtKVazbaNjc5tKSaebRB3311mNcpkPv7RubiiO...	wanshafiqsufian@gmail.com	123	1779332586	active
☐				2	test2	\$2y\$10\$5KjMjB0C43ZwLmW2hdnBd.s/n7aNMG3bdArFpi8KuRQ...	test2@test.com	123	1779332606	active
☐				3	Zikry	\$2y\$10\$Gyv/2Ba/rYM9FYk8BxREOI87b5jWpYjzxjHP7GkUw/...	zikry@gmail.com	0123776635	1779360753	active



PROTOTYPE DEMO





THANK YOU

CONTACT US

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✉ childsafecom.com





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QNA



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APPENDIX

SWOT ANALYSIS

Strength

- Works indoors, no GPS needed
- No monthly fees
- Proactive alert system
- Low manufacturing cost

OPPORTUNITIES

- 770 children lost per year in Malaysia
- RM21.4B global TAM
- Government child safety initiatives

WEAKNESSES

- BLE range limited to 10–30m
- Still on Arduino prototype
- No mobile app yet
- Junior team, 2 years experience

THREATS

- Apple AirTag, Samsung SmartTag
- Big tech entering affordable market
- Price-sensitive B40 market
- MCMC certification delays



u-blox neo-6m



BUSINESS MODEL CANVAS





CHILDSAFE 2.0

1. Safe Zone Detection
2. Buzzer Alert
3. Multi-Child Monitoring
4. SOS Emergency Button
5. Tracking History
6. Telegram Notification
7. Rechargeable Device

BUSINESS MODEL



Freemium Plan Individual Plan

RM0

- ✓ Basic Child Care (Scheduled Hours)
- ✓ Limited Daily Summary Report.
- ✓ Play-Based Activities (General)
- ✓ In app ads

Premium Plan Individual Plan

RM7.99 / month

- ✓ Basic Child Care (Scheduled Hours)
- ✓ Unlimited Daily Summary Reports.
- ✓ Guaranteed Full-Time Spot
- ✓ Ad-free communication portal (if applicable)
- ✓ Specialized Educational Enrichment

Yearly Plan Individual Plan

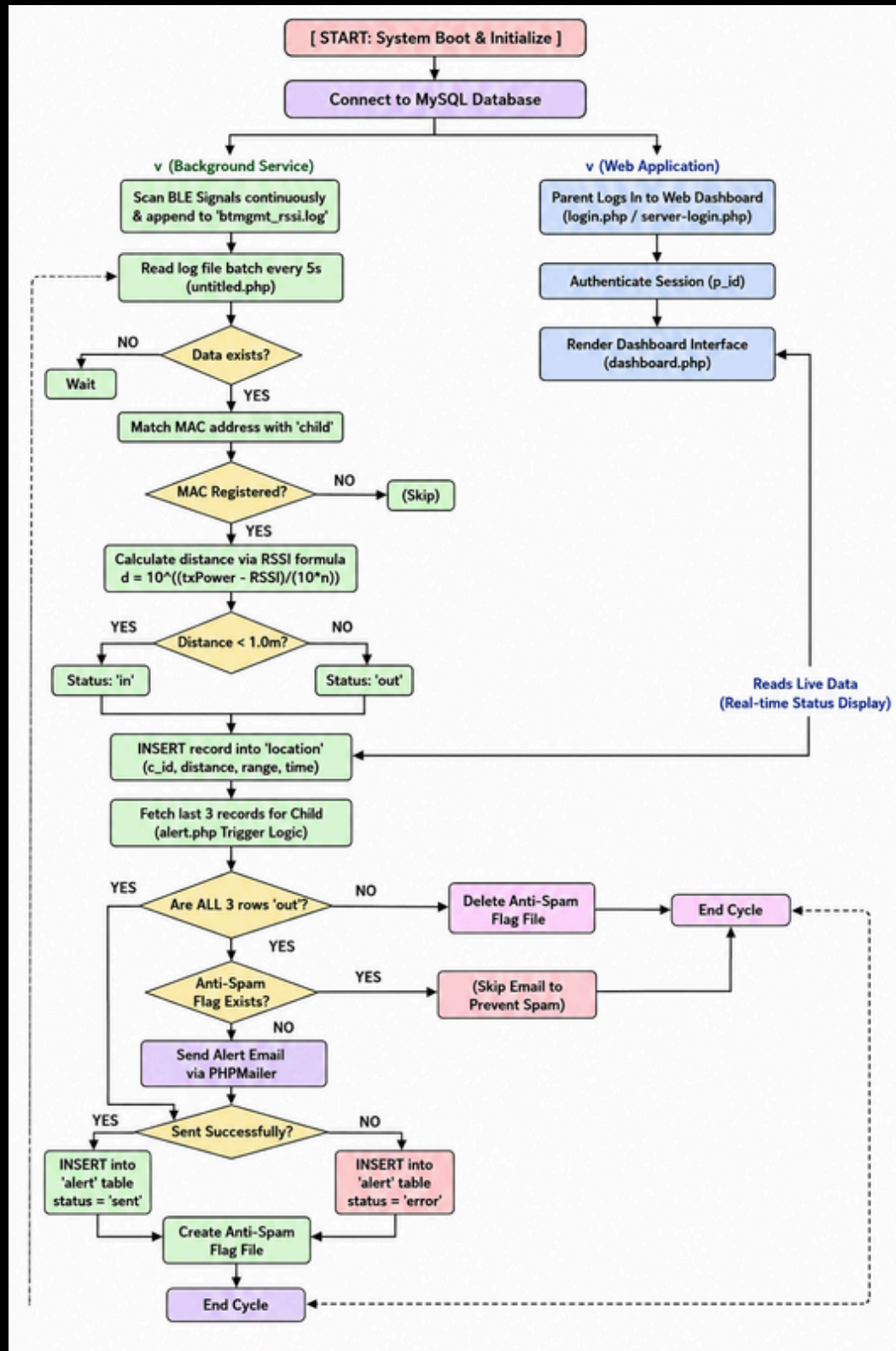
RM20.99 / year

- ✓ Child Care (Extended Hours)
- ✓ Unlimited Daily Summary Reports.
- ✓ Guaranteed Priority Placement
- ✓ Personalized Development Path (Individualized)
- ✓ All Educational Enrichment Programs
- ✓ Parent-Teacher Conferences (Bi-annual)

BUSINESS MODEL



FLOWCHART



To estimate distance from RSSI (Received Signal Strength Indicator), you can use the **Log-Distance Path Loss Model**. This equation estimates distance based on signal degradation, measured in decibels-milliwatts (dBm). [Stack Overflow +4](#)

The formula used to calculate distance (d) is:

$$d = 10^{\frac{\text{Measured Power} - \text{RSSI}}{10 \times n}}$$

```
// Optional: Print to console so you can watch live progress
printf("%s %s\n", time_str, clean_line_start);

// --- Live Real-Time Calculation Block ---
char *rssi_ptr = strstr(clean_line_start, "RSSI:");
if (rssi_ptr != NULL) {
    int val = 0;
    if (sscanf(rssi_ptr, "RSSI: %d", &val) == 1) {
        double txPower = -24.5;
        double n = 4.0;
        double distance_meters = pow(10.0, (txPower - (double)val) / (10.0 * n));
        double distance_cm = distance_meters * 100.0;

        printf("    -> Raw RSSI: %d dBm | Distance: %.2f cm | Status: %s\n",
               val, distance_cm, (distance_cm < 100.0) ? "In range" : "Out of range!");
    }
}
pclose(fp);
sleep(2); // Safeguard cooldown before auto-restarting pipeline if dropped
```

- $n = 2 \rightarrow$ Free space / open air
- $n = 2.7 - 3.5 \rightarrow$ Urban area outdoors
- $n = 4.0 - 6.0 \rightarrow$ Indoor with obstacles (walls, people)

FILE SYSTEM



Log file

```
Player | [Pause] [Full Screen] [Refresh] [Close]
Applications | Mozilla Firefox | untitled.c - /home/traini... | btmgmt_rssi.log (~/Gro... | training@ubuntu: ~/Gro... | Group4
btmgmt_rssi.log (~/Group4) - gedit
btmgmt_rssi.log
~/Group4

The file "/home/training/Group4/btmgmt_rssi.log" changed on disk.

1 [2026-05-21 15:31:06] Device C4:BE:84:E1:1F:F9 RSSI: -68
2 [2026-05-21 15:31:13] Device C4:BE:84:E1:1F:F9 RSSI: -58
3 [2026-05-21 15:31:13] Device C4:BE:84:E1:1F:F9 RSSI: -70
4 [2026-05-21 15:31:13] Device C4:BE:84:E1:1F:F9 RSSI: -60
5 [2026-05-21 15:31:13] Device C4:BE:84:E1:1F:F9 RSSI: -74
6 [2026-05-21 15:31:13] Device C4:BE:84:E1:1F:F9 RSSI: -60
7 [2026-05-21 15:31:29] Device C4:BE:84:E1:1F:F9 RSSI: -68
8 [2026-05-21 15:31:29] Device C4:BE:84:E1:1F:F9 RSSI: -58
9 [2026-05-21 15:31:43] Device C4:BE:84:E1:1F:F9 RSSI: -50
10 [2026-05-21 15:31:43] Device C4:BE:84:E1:1F:F9 RSSI: -60
11 [2026-05-21 15:32:04] Device C4:BE:84:E1:1F:F9 RSSI: -52
12 [2026-05-21 15:32:04] Device C4:BE:84:E1:1F:F9 RSSI: -62
13 [2026-05-21 15:32:04] Device C4:BE:84:E1:1F:F9 RSSI: -54
14 [2026-05-21 15:32:17] Device C4:BE:84:E1:1F:F9 RSSI: -64
15 [2026-05-21 15:32:17] Device C4:BE:84:E1:1F:F9 RSSI: -56
16 [2026-05-21 15:33:57] Device C4:BE:84:E1:1F:F9 RSSI: -64
17 [2026-05-21 15:33:57] Device C4:BE:84:E1:1F:F9 RSSI: -56
18 [2026-05-21 15:34:50] Device C4:BE:84:E1:1F:F9 RSSI: -76
19 [2026-05-21 15:34:50] Device C4:BE:84:E1:1F:F9 RSSI: -58
20 [2026-05-21 15:34:50] Device C4:BE:84:E1:1F:F9 RSSI: -66
21 [2026-05-21 15:34:50] Device C4:BE:84:E1:1F:F9 RSSI: -58
22 [2026-05-21 15:34:50] Device C4:BE:84:E1:1F:F9 RSSI: -72
```


BATERI LIFE



Theory

1. Power Bank Technical Specifications (From Image)

- Internal Cell Capacity: 20,000 mAh (74Wh)
- Rated Capacity (at 5V output): 12,500 mAh (This is the actual capacity delivered to the Arduino after the voltage conversion process to 5V).
- Current Battery Level: 99%

Therefore, the currently available energy capacity is:

$$\text{Available Capacity} = 12,500 \text{ mAh} \times 0.99 = \mathbf{12,375 \text{ mAh}}$$

3. Runtime Formula & Calculations (At 99% Battery)

$$\text{Runtime (Hours)} = \frac{\text{Available Capacity (mAh)}}{\text{Circuit Current (mA)}}$$

$$\text{Runtime (Hours)} = \frac{12,375 \text{ mAh}}{90 \text{ mA}} = \mathbf{137.5 \text{ Hours}}$$

Converting hours into days (for continuous 24-hour operation):

$$\text{Runtime (Days)} = \frac{137.5 \text{ Hours}}{24 \text{ Hours}} = \mathbf{5.73 \text{ Days}}$$

4. Alternative Scenario: Operating for Only 6 Hours per Day

If you only turn on your project for 6 hours each day instead of leaving it running 24/7, the math is:

$$\text{Runtime (Days)} = \frac{137.5 \text{ Total Hours}}{6 \text{ Hours per Day}} = \mathbf{22.91 \text{ Days}}$$

BATERI LIFE



Real-World Expectation

Slide Title: The Reality of Battery Percentage (100% to 99%)

- **The Observation:** * The power bank was used since yesterday, but the battery only dropped from 100% to 99%.
 - *Initial Assumption:* The battery drops at a rate of 1% per day (lasting 100 days).
- **The Reality (Software & Hardware "Trick"):**
 - **Over-saturation at 100%:** When fully charged, the battery cells hold maximum peak voltage. The digital display stays "stuck" at 100% or 99% much longer than any other percentage.
 - **Non-Linear Discharge Curve:** Lithium batteries do not drain at a perfectly steady, even rate.
- **How the Battery Actually Drains:**
 - **Top Tier (100% – 90%):** Drops extremely slowly (the current phase being observed).
 - **Middle Tier (90% – 20%):** The percentage will start dropping much faster and more accurately based on the Arduino's 90 mA current draw.
 - **Bottom Tier (Below 20%):** Plummets very rapidly to 0%.

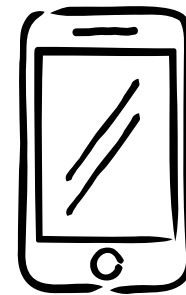


CHILD SAFE

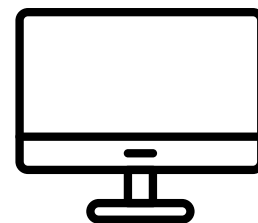
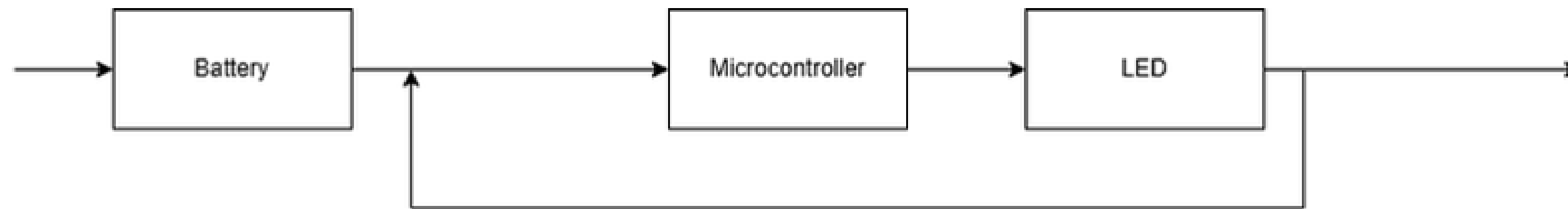
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HIGH LEVEL BLOCK DIAGRAM



Device A



Device B

